**Bogleheads**

# Approximating Vanguard target date funds

**Approximating Vanguard target date funds** aims to help you if you are saving and investing in an employer provided plan and want to approximate a Vanguard target date fund based on the funds in your plan.

If you are lucky enough to have an employer plan with a wide selection of low-cost asset class index funds, you can probably closely approximate a target date fund with a limited number of funds.

However, you may find that your plan has high costs, offering a lot of funds but few of which are index funds. For you, creating a simple, low cost approximation can be more difficult. You might need to use other accounts to reach the asset allocation you are aiming for.

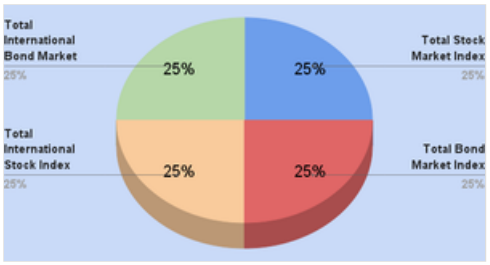
This article describes the how to approximate a target date fund, and include a comprehensive example. It also shows several cases where approximating target retirement funds is suitable.

## Principles of approximating Vanguard target date funds

The Vanguard target date retirement funds are **four-fund portfolios.** The four asset class index funds making up this buy-hold-and rebalance portfolio are:

- A US total stock market index fund
- A total international stock market index fund
- A US total bond market index fund
- A total international bond market index fund

The Vanguard Target Retirement Income fund, as well as the funds that are closing in on their target dates add a fifth fund, the Vanguard Short-Term Inflation Indexed Bond Index.



Four-fund portfolio[note 1]

Target date funds change allocations over time. If you approximate a target retirement fund you will need to **change allocations regularly** to follow changes in the target retirement fund you are approximating. See table Vanguard target retirement funds - Detailed fund allocations.

Before Vanguard added international bonds, target funds held the three-fund portfolio without international bonds. If you want to approximate a target retirement fund, but you do not want to hold international bonds, you can replace these international bonds with domestic bonds.

## Constituent total market funds

The three-fund and four-fund portfolios have one thing in common: Every index fund name contains "total." This section shows the "total" index fund compositions.

### US total stock market funds

These funds invest in the investable US public stock markets, weighting stocks according to their market capitalization. They will include large-cap, mid-cap, and small-cap stocks. A style box (https://www.morningstar.com/content/dam/marketing/apac/au/pdfs/Legal/Stylebox_Factsheet.pdf) shows the percentage of stocks in each weighting category. Summing all categories equals 100%. The weighting of these components can change over time. Current Vanguard and Fidelity total US stock market fund weightings are available here (https://www.morningstar.com/funds/xnas/vtsax/portfolio) and here (https://www.morningstar.com/funds/xnas/fskax/portfolio).

Fund style box (%)[note 2]

| Value | Blend | Growth | |
|---|---|---|---|
| 26 | 22 | 24 | **Large** |
| 6 | 6 | 7 | **Mid** |
| 3 | 3 | 3 | **Small** |

#### Approximating US total stock market funds

*Main article: Approximating total stock market*

It is likely that you will find only an S&P 500 index fund in your retirement plan. In this case, for total market coverage you need to add a completion index fund, or a combination of mid cap and small cap index funds.

In some cases, you may have these funds in your plan. The Thrift Savings Plan, the federal workers retirement savings plan, for example, offers both an S&P 500 index fund and an extended market completion index fund. Some plans may also offer a mid cap index fund and a small cap index fund in addition to the S&P 500 index fund, which makes it possible to approximate the US total market.

#### Without suitable options for US total stock market funds

If your plan lacks suitable options to capture the rest of market, you may be able to do this by:

- Investing in these completion indexes in a personal retirement plan or taxable account;
- Forgo investing in the S&P 500 fund and hold the total market fund allocation in a personal retirement plan or taxable account.
- If married, invest in these completion indexes in your spouse's accounts.
- If married, forgo investing in the S&P 500 fund and hold the total market fund allocation in one of your spouse's accounts.

### Total international stock market index fund

The most complete total international index funds invest in both developed market and emerging market stocks, and hold market allocations to large-cap, mid-cap, and small-cap stocks.

### Approximating total international stock market index fund

*Main article: Approximating total international stock market*

Some international index funds will only cover developed markets, and many international index funds only include large-cap and mid-cap stocks. In these cases, you can consider adding emerging market and small cap international index funds. Depending on the specific indexes used for developed markets and emerging markets, completely replicating the total market could mean that you need to add Canadian and South Korean index funds.

The approximate breakdown between developed market, emerging market, and global-ex US small cap stocks is:[1]

- Developed market: 65%
- Emerging market: 20%
- Global ex-US small cap: 15%

Depending on the availability of emerging market and small cap international index funds, you might be able to add them in your employer's plan, in your individual retirement plans or taxable accounts, or if married, in your spouse's accounts. An additional option is to forgo investing in a developed market index fund and hold a total international index fund in another account (if married, this can also include a spouse's account).

### Total US bond index funds

In the US, a total bond index is commonly benchmarked to the Bloomberg US Aggregate Bond Index, which benchmarks investment grade taxable bonds. These include treasury and government agency bonds, investment grade corporate bonds, mortgage-backed securities and asset-backed securities. The index does not include Treasury inflation protected securities, high yield bonds, floating rate bonds, or municipal bonds. The total US bond market index has an intermediate term duration.

### Total international bond index funds

Most forum investors will be considering the Vanguard international bond index fund, which is hedged against currencies, and invests in international government, agency, and corporate bonds, primarily from developed countries. The index has an intermediate term duration.

## Example of approximating a Vanguard target date fund

Below is an illustrative example, as of August 31, 2013, which builds a target date fund asset allocation from a 401(k) plan's list of available funds.[note 3] The plan has:

- A lot of choices.
- Target retirement funds, but in this case these are inappropriate because they are Fidelity Freedom® Funds (https://www.fidelity.com/mutual-funds/asset-allocation-funds/freedom-funds/overview) that use 24 active funds.[note 4]

- A Balanced fund, Fidelity® Multi-Asset Index Fund (https://fundresearch.fidelity.com/mutual-funds/summary/31634R109), which approximates three quarters of the Vanguard target retirement fund (lacks international bonds) but comes with the limitation of having only one asset allocation (85% stocks/ 14% bonds).
- U.S. and international stock total market indexes and a US bond index that match three quarters of the Vanguard target retirement fund and fulfill the requirements for a three-fund portfolio.
- A developed international market index.
- Additional bond indexes.

The table below shows the complete list of fund choices and expense ratios:

| 401(k) plan options | |
|---|---|
| **Funds** | **Expense ratio** |
| **Target retirement funds** | |
| FID FREEDOM K 2000 (FFKBX) | 0.39% |
| FID FREEDOM K INCOME (FFKAX) | 0.39% |
| FID FREEDOM K 2005 (FFKVX) | 0.46% |
| FID FREEDOM K 2010 (FFKCX) | 0.51% |
| FID FREEDOM K 2015 (FKVFX) | 0.52% |
| FID FREEDOM K 2020 (FFKDX) | 0.55% |
| FID FREEDOM K 2025 (FKTWX) | 0.60% |
| FID FREEDOM K 2030 (FFKEX) | 0.62% |
| FID FREEDOM K 2035 (FKTHX) | 0.66% |
| FID FREEDOM K 2040 (FFKFX) | 0.66% |
| FID FREEDOM K 2045 (FFKGX) | 0.68% |
| FID FREEDOM K 2050 (FFKHX) | 0.68% |
| FID FREEDOM K 2055 (FDENX) | 0.69% |
| **Balanced funds** | |
| FID FOUR IN ONE IDX (FFNOX) | 0.24% |
| PIM ALL ASSET INST (PAAIX) | 0.98% |
| **US stock funds** | |
| SPTN TOTAL MKT IDX ADV (FSTVX) | 0.07% |
| SPTN 500 INDEX INST (FXSIX) | 0.05% |
| SPTN MID CAP IDX ADV (FSCKX) | 0.22% |
| VANG SM CAP IDX INST (VSCIX) | 0.08% |
| SPTN EXTENDED MKT IDX ADV (FSEVX) | 0.07% |
| DODGE & COX STOCK (DODGX) | 0.52% |
| FID GROWTH CO K (FGCKX) | 0.77% |
| SPTN REAL ESTATE IDX ADV (FSRVX) | 0.20% |
| **International stock funds** | |
| AF EUROPAC GROWTH R5 (RERFX) | 0.55% |
| NB INTL EQUITY TRUST (NIQTX) | 1.57% |
| SPTN INTERNATIONAL INDEX ADV (FSIVX) | 0.17% |
| SPTN GLB XUS IDX ADV (FSGDX) | 0.28% |
| **US bond funds** | |
| FID SEL MONEY MARKET (FSLXX) | 0.30% |
| SPTN SHORT TERM TREASURY BOND IDX ADV (FSBAX) | 0.10% |
| SPTN INFLATION PROTECTED Bond IDX ADV (FSIYX) | 0.10% |
| SPTN INTERMEDIATE TREASURY Bond IDX ADV (FIBAX) | 0.10% |
| SPTN LONGTERM TREASURY BOND IDX ADV (FLBAX) | 0.10% |
| SPTN US BOND IDX ADV (FSITX) | 0.17% |

## Goal

The following examples use an asset allocation of 60% / 40% stocks to bonds. By design, Vanguard allocates 28% of a target date fund's stock allocation to international stocks and allocates 20% of a portfolio's bond allocation to international bonds. The Vanguard Target Retirement 2020 Fund (VTWNX) (https://investor.vanguard.com/investment-products/mutual-funds/profile/vtwnx) is the closest match and holds the following total market index funds:[note 5]

**Vanguard Target Retirement 2020 Fund (VTWNX), as of August 31, 2013**

| Ranking by Percentage | Fund | Percentage | Stocks | Bonds |
|---|---|---|---|---|
| 1 | Vanguard Total Stock Market Index Fund Investor Shares | 43.6% | 43.6% | |
| 2 | Vanguard Total Bond Market II Index Fund Investor Shares | 30.3% | | 30.3% |
| 3 | Vanguard Total International Stock Index Fund Investor Shares | 18.5% | 18.5% | |
| 4 | Vanguard Total International Bond Index Fund Investor Shares | 7.6% | | 7.6% |
| Total | — | 100.0% | 62% | 38% |

The last two columns show which funds are categorized as stocks or bonds. Asset allocations only need to be within 5%, so this target retirement fund is suitable for a 60 / 40 portfolio.

The following Google spreadsheet documents the scenarios below:

**Approximating a Vanguard Target Date Fund**

Approximating a Vanguard target date fund : Sheet1

| Funds | Expense ratio | Vanguard's fund list: | https://personal.vanguard.com/us/fu |
|---|---|---|---|
| **Target retirement funds** | | | |
| FID FREEDOM K 2000 (FFKBX) | 0.39% | | |
| FID FREEDOM K INCOME (FFKAX) | 0.39% | | |
| FID FREEDOM K 2005 (FFKVX) | 0.46% | | **Vanguard Target Retirement 2020** |
| FID FREEDOM K 2010 (FFKCX) | 0.51% | | https://personal.vanguard.com/us/fu |
| FID FREEDOM K 2015 (FKVFX) | 0.52% | **Ranking by Percentage** | **Fund** |
| FID FREEDOM K 2020 (FFKDX) | 0.55% | 1 | Vanguard Total Stock Market Index F |
| FID FREEDOM K 2025 (FKTWX) | 0.60% | 2 | Vanguard Total Bond Market II Index |
| FID FREEDOM K 2030 (FFKEX) | 0.62% | 3 | Vanguard Total International Stock Ir |

Sheet1 < >

(View Google Spreadsheet in browser (http://spreadsheets.google.com/ccc?key=0AsVSNcUR-B6IdHJReXNoMVRVTHI0UTBtTVBEMEhrR3c), then File --> *Download as* to download the file.)

Note: If the spreadsheet is blank, select a different sheet, then back to that sheet. The image will be refreshed.

## Approximating the U.S stock market allocation

This plan holds a total U.S. stock market index fund. Replace the Vanguard funds as shown in the spreadsheet under "A total US stock market fund is available." Because it lacks a total international bond market index, one option is to use the US bond index fund in place of the international bond index fund. Round the asset allocations to the nearest 5%.

The following scenarios show how to approximate the U.S. total market segment, if we assume that the plan *did not provide* a total U.S. stock market index.

### No U.S. total stock market index fund is available (completion index)

If the plan lacks a U.S. total stock market index fund, you can use the available completion index fund and reconstitute the total US market slice of the portfolio as follows:

S&P 500 index fund with completion index fund. The total market approximation split would be:

- S&P 500 index: 80% of the US stock allocation
- Completion index: 20% of the US stock allocation.

Refer to in the spreadsheet under "If a total US stock market index fund is not available (Completion index)." In this case, entering FXSIX (holding value of 80%) and FSEVX (holding value of 20%) will result in a Total US Stock style box that's very close to the one shown above.

Split the US total stock market index share by these percentages as shown:

- S&P 500 index: 34.8% = 80% of 43.6%
- Completion index: 8.72% = 20% of 43.6%

This example uses extra precision (more decimal places) to show the methodology. In practice you round the final asset allocations to the nearest 5%. Rounding the example percentages gives this allocation:

- S&P 500 index: 35%
- Completion index: 10%

#### No U.S. total stock market index fund is available (small-cap and mid-cap)

If the plan lacks both the U.S. total stock market index and a completion index fund, you could use the available small-cap and mid-cap index funds to approximate the US total market index:

S&P 500 index fund with a mid-cap index fund and a small-cap index fund. The total market approximation split would be:

- S&P 500: 80% of the US stock allocation
- Mid cap index: 5% of the US stock allocation
- Small cap index: 15% of the US stock allocation.

Refer to in the spreadsheet under "If a total US stock market index fund is not available (Mid-cap, small-cap)." As stated previously, use Morningstar's Instant X-Ray.[note 6] In this case, entering FXSIX (holding value of 80%), FSCKX (holding value of 5%), and VSCIX (holding value of 15%) will result in a Total US Stock style box that's very close to the one shown above.

Split the US total stock market index share by these percentages as shown:

- S&P 500: 34.88 % = 80% of 43.6%
- Mid cap index: 2.18 % = 5% of 43.6%
- Small cap index: 6.54% = 15% of 43.6%

This example uses extra precision (more decimal places) to show the methodology. In practice you round the final asset allocations to the nearest 5%. Rounding the example percentages gives this allocation:

- S&P 500: 35%
- Mid cap index: 5%
- Small cap index: 5%

#### No U.S. total stock market index fund is available (S&P 500 index only)

The plan only has an S&P 500 index fund and neither of the above funds are available. In this case, you would need to use a completion index fund, or a combination of small-cap and mid-cap index funds held in *other* accounts (personal retirement accounts, a taxable account, or if married, one of your spouse's accounts) to approximate a total market position.

Otherwise, use the S&P 500 index fund.

In addition, you might decide to totally forgo investing in the plan's US stock funds, by holding the US stock allocation in another account (or if married, in your spouse's account.)

### Approximating the international stock allocation

This plan holds a total international stock market index fund; which is shown in the spreadsheet under "A total US stock market fund is available."[note 7]

The following scenario shows how to approximate the international total market segment if the plan *did not provide* a total international index fund.

#### No total international stock market index fund is available

The plan has developed market index fund (SPTN International Index ADV (FSIVX)).

To partially approximate the total international fund, you would invest in the developed market index fund (FSIVX) but would need to add an emerging market index fund to the portfolio in another account. For a closer approximation, you would also need to add a global-ex U.S. small cap international index fund in another account. In this particular scenario, because the plan's developed market index tracks the MSCI EAFE index, the allocation lacks exposure to Canada. If the added emerging market index does not include South Korea, this nation would also be missing from the allocation.[note 8]

In addition, you might decide to totally forgo investing in the plan's international stock funds, by holding the international allocation in another account (or if married, in your spouse's account.)

### Approximating the U.S. bond allocation

This plan holds a U.S. total bond market index fund; which is shown in the spreadsheet under "A total US stock market fund is available."

The following scenario shows how to approximate the U.S. total bond market index, if we assume that the plan *did not provide* a total U.S bond index.

Refer to Inflation-protected securities for suggestions on incorporating inflation-protected bonds.

#### No U.S. total bond market index fund is available

The plan has a U.S. intermediate-term Treasury bond index fund available that provides the portfolio with an investment grade intermediate bond allocation. If you want to diversify the bond allocation, you can do it by holding additional bond funds in other accounts.[note 9]

In addition, you might decide to totally forgo investing in the plan's US bond funds, by holding the US bond allocation in another account (or if married, in your spouse's account.)

### Approximating the international bond allocation

This plan does not provide an international bond allocation. You need to add an international bond allocation in another account.

Alternatively, if you do not want to hold international bonds, you could fill this allocation with a U.S. total bond index, U.S. Treasuries, Inflation-protected securities or, in taxable accounts, tax-exempt bonds.

## Notes

1. The above graphic represents that there are four primary components to a target date fund. It does not indicate that a typical allocation is 25% in each.
2. The weightings shown here are for illustration only.
3. See Bogleheads forum topic: "First 401(k) Advice" (https://www.bogleheads.org/forum/viewtopic.php?t=122977). Disclaimer: The examples have no relation to the member's situation and are only illustrative.
4. Actively managed target retirement funds often are a compilation of a large collection of actively managed funds. Here are the fund tallies from four large mutual fund complexes for target date 2035 funds:
   - T. Rowe Price Retirement 2035 Fund (https://www3.troweprice.com/fb2/fbkweb/composition.do?ticker=TRRJX), from T. Rowe Price (https://individual.troweprice.com/public/Retail): 18 funds (one index fund).
   - Fidelity Freedom® 2035 Fund (https://fundresearch.fidelity.com/mutual-funds/summary/315792655), from Fidelity (https://www.fidelity.com/): 24 funds.
   - American Funds 2035 Target Date Retirement Fund® (https://americanfundsretirement.retire.americanfunds.com/about/funds/details.htm?ticker=AAFTX), from American Funds (https://americanfundsretirement.retire.americanfunds.com/home.htm): 17 funds
   - One Choice 2035 Portfolio (https://ipro.americancentury.com/ipro/product/mf/summary/954.jsp#ARYIX), from American Century Investments (https://ipro.americancentury.com/ipro/index.jsp): 13 funds.
5. The allocations below represent the Vanguard 2020 fund as they were in 2013. The allocations as of today have significantly changed, as indicated in the above link, with a lower stock allocation and a higher bond allocation.
6. Morningstar's *Instant X-Ray* portfolio analysis tool has been discontinued effective April 17, 2025. Source: "Upcoming changes to our membership offerings, tools, and features" (https://global.morningstar.com/en-gb/updates/upcoming-changes-our-membership-offerings-tools-features). *Morningstar*. April 15, 2025. Retrieved March 10, 2026. Forum member LadyGeek has developed an alternative Equity style box calculator (https://docs.google.com/spreadsheets/d/113HpxXv5NrKWDXpv_xGNIVeOQk3giuRD/edit?pli=1&gid=63194195#gid=63194195) spreadsheet. Discussion and support is in Bogleheads forum topic: "Wiki - Morningstar's free Instant X-ray tool no longer available - Alternatives?" (https://www.bogleheads.org/forum/viewtopic.php?t=469099), LadyGeek. March 08, 2026.
7. Note that the plan's total international index fund tracks the MSCI All-World ex-US index, which covers the world's large-cap and mid-cap stocks. To more completely approximate the Vanguard target date fund allocations, you would need to add a global ex-US small cap fund.

8. For a precise replication of a Vanguard retirement date fund allocation to total international markets, see Approximating Total International Stock Market
9. If you must hold the total bond market index fund allocation in a taxable account, consider substituting a low-cost intermediate tax-exempt bond fund if this gives you better after-tax returns.

## See also

- Vanguard target retirement funds
- Approximating total international stock market
- Approximating total stock market
- How to build a lazy portfolio

## References

1. "Vanguard Total International Fund" (https://institutional.vanguard.com/investments/product-details/fund/0113). Vanguard.

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